

Job Description

Position: Manager/Area Manager Artificial Intelligence position at Tata Steel.

The Context / Purpose of the Job: TSL is seeking an AI Engineer to lead our advancement toward Industry 5.0 by developing and implementing a comprehensive AI strategy focused on creating innovative operating models that seamlessly integrate human and machine intelligence. This key role is centered on writing code, analyzing data, and building robust ML and AI pipelines to drive significant improvements in business performance.

The candidate should be proficient in Python, data science, ML (Time series, optimisation, classification & regression problems), and deep learning modules (CNN, RNN, LSTM), as well as cutting-edge Generative AI concepts like Agentic Design, RAG, prompt-to-template strategies, and chunking strategies.

This position requires expertise in decentralized architectures, proficiency with SQL, NoSQL, and PostgreSQL databases, and the ability to deploy solutions across multi-cloud platforms (GCP, AWS, Azure), leveraging serverless technologies like Cloud Run, Cloud Functions, and Lambda, alongside Kubernetes for container orchestration.

Key Objective / Overall Job Responsibility (Main Purpose):

- Strategic AI Vision & Leadership: Develop and champion a clear AI vision by conceptualizing autonomous business models that leverage Generative AI, including Agentic Design and RAG architectures, to redefine the TSL value chain. Collaborate with stakeholders to define solution objectives, deliverables, and timelines.
- Hands-On AI Implementation & Deployment: Lead the end-to-end design, development, and deployment of AI solutions using Python and its data science ecosystem. Architect and build robust ML/AI pipelines and deploy them on multi-cloud serverless platforms (GCP Cloud Run, AWS Lambda, Azure Functions) and Kubernetes (GKE).
- Advanced Model Development & Innovation: Direct the AI learning trajectory by applying Deep Learning models (CNN, RNN, LSTM) and advanced Generative AI techniques, including innovative chunking strategies and prompt-to-template frameworks, to solve complex business challenges.
- Technical Leadership & Data Architecture: Provide technical guidance to internal and external partners to build scalable AI solutions on decentralized architectures. Ensure seamless implementation and data integrity through proficient use of SQL (PostgreSQL) and NoSQL databases.
- Program & Change Management: Spearhead the end-to-end transformation process, from model conception to production deployment, while managing change to ensure successful business adoption. Effectively multitask across multiple AI projects, prioritizing technical resources to meet competing deadlines.
- Communication & Stakeholder Engagement: Effectively translate and communicate complex model performance metrics, data-driven insights, and strategic project direction to diverse audiences, including senior management, business users, and external partners, to ensure alignment and buy-in.

Relevant Experience:

- Experience of working in high performance teams delivering AI solutions
- Experience of working in cross-functional & collaborative environment

- Good understanding of mining / manufacturing / supply chain / commercial processes along with knowledge of technology applications in these domains, would be preferred

Skills/Technical Competencies:

- Deep understanding of AI/ML algorithms and techniques, including Generative AI models (e.g., large language models, SLM, diffusion models), supervised, unsupervised, and reinforcement learning.
- Expertise in AI architecture design and implementation: Familiarity with cloud platforms (AWS, Azure, GCP), big data technologies (Hadoop, Spark), and AI/ML frameworks (TensorFlow, PyTorch).
- Understanding of frameworks & methods like RAG, Fine-tuning, Pre-training, Agentic AI
- Proficiency in Prompt Engineering, utilizing tools, accessing APIs, and collaborating with AI agents
- In-depth understanding of the LangChain, Google ADK etc framework for building and optimizing Large Language Model (LLM) applications.
- Experience with data engineering and data management, including data cleaning, pre-processing, feature engineering, and data pipelines.
- Proficiency in programming languages: Python, R, or similar languages are essential.
- Knowledge of model deployment and monitoring: Experience with MLOps practices and tools for deploying and managing AI models in production environments.
- Understanding of bias mitigation, privacy preservation, and responsible AI practices.
- Familiarity with various AI applications and their business implications, across different functional areas.
- Ability to design scalable, robust, and maintainable AI systems.
- Understanding of IT-OT architecture

Behavioural Competencies:

- Proficiency in forging strong Customer, Supplier, Partner relationships
- Managing multiple AI projects simultaneously, ensuring timely and successful delivery within budget.
- Target & timeline oriented
- Effectively communicating technical concepts to both technical and non-technical stakeholders, managing expectations, and building consensus.
- Understanding the business context and translating business needs into technical solutions.

Educational qualifications: BE/BTech, ME/MTech, MSc (Maths / Stats), MBA/PGDM (equivalent) from distinguished Institutions